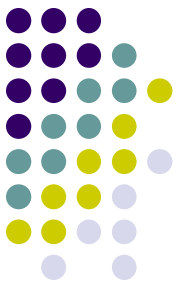


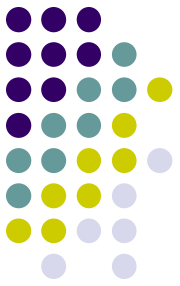
Chapter 12. Models for Cross Classified Random Effects

An Example: Students jointly nested in schools and communities (Scotland data)



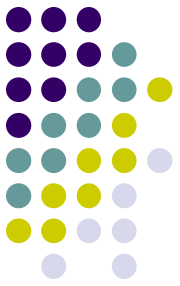
Neighborhood	School																	Totals
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
26																	5	5
27																	1	1
29															1		8	9
30																	2	2
31															1		1	2
32															1		5	6
33															1		2	3
35																	3	3
36																	2	2
38															1		4	5
.																		.
.																		.
.																		.
251														4		1		5
252													1	3			1	5
253														3				3
256																	2	2
258												4						4
259											6	1					2	9
260											6							6
261											3						3	6
263											14					1	1	16
.																		.
.																		.
.																		.
793									6									6
794			1			1			11									13
795				2					1									3
796				9					1									10
797				3				1										4
798								10										10
799			1							1								2
800			3	1														4
801			1	1														2
803										3								3
Totals	146	22	146	159	155	101	286	112	136	133	92	190	111	154	91	102	174	2,310

Possible Analysis Goals:



- Components of variance between neighborhoods, between schools and among children (within the “cells”)
- Student, neighborhood and school predictors of student outcomes
- Assess variation in association between student predictors and outcomes over neighborhoods, schools and cells
- Assess variation in neighborhood effects over schools
- Assess variation in school effects over neighborhood
- Estimate unique random effects for schools and neighborhoods

Cross Random Effects Model



Level-1 or “Within-Cell” Model (unconditional model)

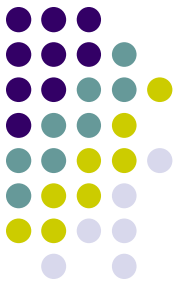
$$Y_{ijk} = \pi_{0jk} + e_{ijk}, \quad e_{ijk} \sim N(0, \sigma^2)$$

where

Y_{ijk} is the outcome for student i , in neighborhood j (rows), and school k (columns)

π_{0jk} is the mean attainment of children in cell jk ; i.e. who live in neighborhood j and attend school k

e_{ijk} is the random “child effect”



Level-2, or “Between-Cell” Model

$$\pi_{0jk} = \theta_0 + b_{00j} + c_{00k} + d_{0jk}$$

$$b_{00j} \sim N(0, \tau_{b00})$$

$$c_{00k} \sim N(0, \tau_{c00})$$

$$d_{0jk} \sim N(0, \tau_{d00})$$

where

θ_0 is the grand-mean achievement of all children;

b_{00j} is the random main effect of neighborhood j averaged over all schools (rows)

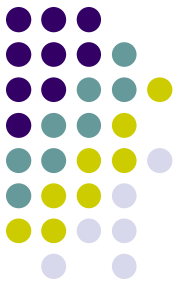
c_{00k} is the random main effect of school k averaged over all neighborhoods (columns)

d_{0jk} is the interaction effect

Combined Model

$$Y_{ijk} = \theta_0 + b_{00j} + c_{00k} + d_{0jk} + e_{ijk}$$

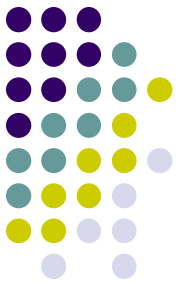
a two-way ANOVA with random row effects, column effects, two-way interactions, and within-cell deviations.



Variance Partitioning

- within-cells, σ^2
- between-cells ---- three components:
 - variance between neighborhoods, τ_{b00}
 - variance between schools, τ_{c00}
 - residual variance between cells, τ_{d00}
(not estimable in many applications; confounded with σ^2)

Three Kinds of Intra-unit Correlation Coefficients



1. Outcomes of 2 children in the same neighborhood and the same school:

$$\text{corr}(Y_{ijk}, Y_{i'jk}) = \rho_{bcd} = \frac{\tau_{b00} + \tau_{c00} + \tau_{d00}}{\tau_{b00} + \tau_{c00} + \tau_{d00} + \sigma^2}$$

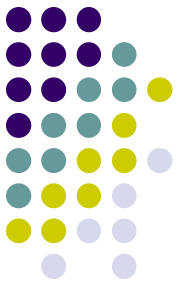
2. Outcomes of 2 children in the same neighborhood but attend different schools:

$$\text{corr}(Y_{ijk}, Y_{i'jk'}) = \rho_b = \frac{\tau_{b00}}{\tau_{b00} + \tau_{c00} + \tau_{d00} + \sigma^2}$$

3. Outcomes of 2 children in the same school but live in different neighborhoods:

$$\text{corr}(Y_{ijk}, Y_{i'j'k}) = \rho_c = \frac{\tau_{c00}}{\tau_{b00} + \tau_{c00} + \tau_{d00} + \sigma^2}$$

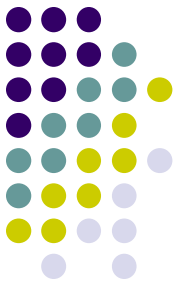
Conditional Models



Each source of variability may be accounted for by school characteristics, neighborhood characteristics, and interactions. For example, a neighborhood characteristic could have effect on the outcome. That neighborhood characteristic could:

- be fixed across schools
- non-randomly vary across schools
- vary randomly across schools

Student-Level



Within-Cell Model (An example)

Suppose the model specifies a cell mean achievement and a gender gap:

$$Y_{ijk} = \pi_{0jk} + \pi_{1jk}a_{ijk} + e_{ijk}, \quad e_{ijk} \sim N(0, \sigma^2)$$

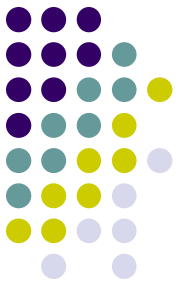
where

π_{0jk} mean achievement in cell jk

π_{1jk} is the regression coefficient relating a_{ijk} to Y_{ijk} ; the “gender gap” in achievement between females and males in cell jk

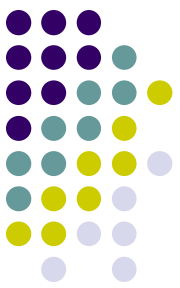
a_{ijk} a student-level predictor of the gender of student ijk (cell-mean centered)

e_{ijk} is a within-cell random effect



Level-2 or “Between-Cell” Model

Level-1 coefficients for mean attainment and gender gap, π_{0jk} and π_{1jk} , become outcomes in the level-2 model that represents between-cell variation created by crossing of the two random factors, neighborhood and school.



(i) Fixed Effects of a Level-2 Row (neighborhood)
Predictor, W_j and a Level-2 Column (school)
Predictor, X_k

$$\pi_{pj k} = \theta_p + \beta_p X_k + \gamma_p W_j + b_{p0j} + c_{p0k} \quad p = 0,1$$

where

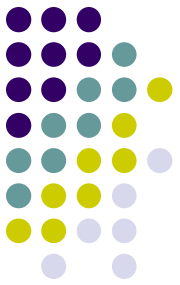
θ_p is the model intercept

β_p is the fixed effect of X_k , a school predictor
(assumed constant over all neighborhoods)

γ_p is the fixed effect of W_j , a neighborhood predictor
(assumed constant over all schools)

b_{p0j}, c_{p0k} , are residual random effects of neighborhoods,
schools

ii) Randomly Varying Effects of a Level-2 Row and a Level-2 Column Predictor



In hierarchical form:

$$\pi_{pjk} = \theta_p + \beta_{pj}X_k + \gamma_{pk}W_j + b_{p0j} + c_{p0k}$$

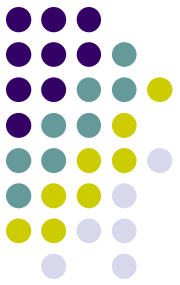
with

$$\beta_{pj} = \beta_p + b_{p1j}$$

$$\gamma_{pk} = \gamma_p + c_{p1k}$$

Notice that β_{pj} and γ_{pk} vary randomly across neighborhoods and schools. Suggests that measured characteristics of the neighborhoods, W_j , might be predictors of β_{pj} . Similarly, measured characteristics of schools, X_k might predict γ_{pk} .

In mixed model form:



Effects of school predictors may vary randomly across neighborhoods;
neighborhood predictors vary randomly across schools.

$$\pi_{pjk} = \theta_p + (\beta_p + b_{p1j})X_k + (\gamma_p + c_{p1k})W_j + b_{p0j} + c_{p0k}$$

Diagram illustrating the mixed model equation:

- School level variable (points to X_k)
- Neighborhood level variable (points to W_j)
- Varies over neighborhood (points to b_{p1j})
- Varies over schools (points to c_{p1k})

β_p is the fixed effect of X_k

b_{p1j} is the random effect of neighborhood j on the association between X_k and π_{pjk}

γ_p is the fixed effect of W_j

c_{p1k} is the random effect of school k on the association between W_j and π_{pjk}

b_{p0j}, c_{p0k} are residual random effects of neighborhoods, schools